

EXERCISE - 1

Q.1 If $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 0 & 5 & 6 \end{bmatrix}$, then $2A =$

[1] $\begin{bmatrix} 2 & 4 & 6 \\ 2 & 3 & 4 \\ 0 & 5 & 6 \end{bmatrix}$

[2] $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 6 & 8 \\ 0 & 5 & 6 \end{bmatrix}$

[3] $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 4 \\ 0 & 10 & 12 \end{bmatrix}$

[4] $\begin{bmatrix} 2 & 4 & 6 \\ 4 & 6 & 8 \\ 0 & 10 & 12 \end{bmatrix}$

Q.2 If $\begin{bmatrix} x & 3x-y \\ zx+z & 3y-w \end{bmatrix} = \begin{bmatrix} 3 & 2 \\ 4 & 7 \end{bmatrix}$, then

[1] $x = 3, y = 7, z = 1, w = 14$

[2] $x = 3, y = -5, z = -1, w = -4$

[3] $x = 3, y = 6, z = 2, w = 7$

[4] None of these

Q.3 If $A = \begin{bmatrix} -2 & 1 \\ 0 & 3 \end{bmatrix}$ and $f(x) = 2x^2 - 3x$, then $f(A)$ equals

[1] $\begin{bmatrix} 14 & 1 \\ 0 & -9 \end{bmatrix}$

[2] $\begin{bmatrix} -14 & 1 \\ 0 & 9 \end{bmatrix}$

[3] $\begin{bmatrix} 14 & -1 \\ 0 & 9 \end{bmatrix}$

[4] $\begin{bmatrix} -14 & -1 \\ 0 & -9 \end{bmatrix}$

Q.4 If $A = \begin{bmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{bmatrix}$ then $A^2 - 5A + 6I =$

[1] $\begin{bmatrix} 1 & -1 & -5 \\ -1 & -1 & 4 \\ -3 & -10 & 4 \end{bmatrix}$

[2] $\begin{bmatrix} 1 & -1 & -3 \\ -1 & -1 & 10 \\ -5 & 4 & 4 \end{bmatrix}$

[3] 0

[4] I

Q.5 If $2A + B = \begin{bmatrix} 6 & 4 \\ 6 & -11 \end{bmatrix}$ and $A - B = \begin{bmatrix} 0 & 2 \\ 6 & 2 \end{bmatrix}$, then $A =$

[1] $\begin{bmatrix} 2 & 2 \\ 4 & -3 \end{bmatrix}$

[2] $\begin{bmatrix} 2 & 0 \\ 4 & -3 \end{bmatrix}$

[3] $\begin{bmatrix} 2 & 2 \\ 4 & 3 \end{bmatrix}$

[4] None of these

Q.6 If $A = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$, then AA^T equals

[1] $\begin{bmatrix} \cos 2\theta & -\sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{bmatrix}$

[2] $\begin{bmatrix} \cos^2 \theta & \sin^2 \theta \\ \sin^2 \theta & \cos^2 \theta \end{bmatrix}$

[3] $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

[4] $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

Q.7 If $A = \begin{bmatrix} 3 & x \\ y & 0 \end{bmatrix}$ and $A = A^T$, then

[1] $x = 0, y = 3$

[2] $x + y = 3$

[3] $x = y$

[4] $x = y$

Q.8 If $A = \begin{bmatrix} \sin^2 \alpha & \sec^2 \alpha \\ \operatorname{cosec}^2 \alpha & 1/2 \end{bmatrix}$ and $B = \begin{bmatrix} \cos^2 \alpha & -\tan^2 \alpha \\ -\cot^2 \alpha & 1/2 \end{bmatrix}$, then $A + B$ is equal to

[1] $\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$

[2] $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

[3] $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

[4] $\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$

Q.9 If $A = \begin{bmatrix} a & b \\ b & a \end{bmatrix}$, then $|A + A^T|$ equals

[1] $4(a^2 - b^2)$

[2] $2(a^2 - b^2)$

[3] $a^2 - b^2$

[4] $4ab$

Q.10 If $A = \begin{bmatrix} 1 & -5 & 7 \\ 0 & 7 & 9 \\ 11 & 8 & 9 \end{bmatrix}$, then the value of $\operatorname{tr} A$ is

[1] 17

[2] 25

[3] 3

[4] 12

Q.11 If $A = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 2 \\ -1 & 1 \end{bmatrix}$, then correct statement is

[1] $AB = BA$

[2] $AA^T = A^2$

[3] $AB = B^2$

[4] None of these

Q.12 $\begin{bmatrix} 1 & -\tan \alpha / 2 \\ \tan \alpha / 2 & 1 \end{bmatrix} \begin{bmatrix} \tan \alpha / 2 & 1 \\ 1 & 0 \end{bmatrix}$ equals

[1] zero matrix

[2] $\sec^2 \alpha \cdot I_2$

[3] I_2

[4] None of these

Q.13 For suitable matrices A, B ; the false statement is

[1] $(AB)^T = A^T B^T$

[2] $(A^T)^T = A$

[3] $(A-B)^T = A^T - B^T$

[4] $(A^T)^{-1} = (A^{-1})^T$

Q.14 For any square matrix A , which statement is wrong

[1] $(\operatorname{adj} A)^{-1} = \operatorname{adj} (A^{-1})$

[2] $(A^T)^{-1} = (A^{-1})^T$

[3] $(A^3)^{-1} = (A^{-1})^3$

[4] None of these

Q.15 $(\operatorname{adj} A^T) - (\operatorname{adj} A)^T$ equals

[1] $2|A|$

[2] $2|A|I$

[3] zero matrix

[4] unit matrix

Q.16 If $A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix}$, then value of A^{-1} is

[1] $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

[2] $\begin{bmatrix} \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{2} & 0 \\ 0 & 0 & \frac{1}{2} \end{bmatrix}$

[3] $\begin{bmatrix} 2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix}$

[4] None of these

Q.17 If $A = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $C = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$, when which of the following statement is true

[1] $AB = BA$

[2] $AB = -BA$

[3] $A^2 = BC$

[4] $A^2 = B + C$

Q.18 Which of the following matrix is inverse of itself

[1] $\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

[2] $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

[3] $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix}$

[4] $\begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

Q.19 If $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, then A^n is equal to ($n \in \mathbb{N}$)-

[1] $\begin{bmatrix} 1 & n \\ 0 & 1 \end{bmatrix}$

[2] $\begin{bmatrix} n & n \\ 0 & n \end{bmatrix}$

[3] $\begin{bmatrix} n & 1 \\ 0 & 1 \end{bmatrix}$

[4] none of these

- Q.20** If $A = \begin{bmatrix} -1 & 0 & 2 \\ 3 & 1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & 5 \\ 2 & 7 \\ 3 & 10 \end{bmatrix}$, then
 [1] AB and BA both exist [2] AB exists but not BA
 [3] BA exists but not AB [4] both AB and BA do not exist
- Q.21** If every row of a matrix A contains p elements and its column contains q elements, then the order of A is
 [1] $p \times p$ [2] $q \times q$ [3] $p \times q$ [4] $q \times p$
- Q.22** If $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$, then
 [1] $AB = BA$ [2] $AB = B^2$ [3] $AB = -BA$ [4] None of these
- Q.23** If A is a matrix of order 3×4 , then both AB^T and $B^T A$ are defined if order of B is
 [1] 3×3 [2] 4×4 [3] 4×3 [4] 3×4
- Q.24** If $A = \begin{bmatrix} 1 & 2 \\ -4 & -1 \end{bmatrix}$, then $A^{-1} =$
 [1] $\frac{1}{7} \begin{bmatrix} 1 & 2 \\ -4 & -1 \end{bmatrix}$ [2] $\begin{bmatrix} -1 & -2 \\ 4 & 1 \end{bmatrix}$ [3] $\frac{1}{9} \begin{bmatrix} 1 & 2 \\ 4 & 1 \end{bmatrix}$ [4] $\frac{1}{7} \begin{bmatrix} -1 & -2 \\ 4 & 1 \end{bmatrix}$
- Q.25** A matrix $A = (a_{ij}) m \times n$ is said to be a square matrix if
 [1] $m = n$ [2] $m \leq n$ [3] $m \geq n$ [4] $m < n$
- Q.26** If order of $A + B$ is $n \times n$, then the order of AB is
 [1] $n \times n$ [2] $n \times m$ [3] $m \times n$ [4] not defined
- Q.27** If A and B are square matrices of same orders, then $\text{adj}(AB)$ equals
 [1] $\text{adj} A \cdot \text{adj} B$ [2] $\text{adj} B \cdot \text{adj} A$ [3] $\text{adj} A + \text{adj} B$ [4] $\text{adj} A - \text{adj} B$
- Q.28** If $A = \begin{bmatrix} 4 & 2 \\ 3 & 3 \end{bmatrix}$, then $\text{adj}(\text{adj} A)$ is equal to
 [1] $\begin{bmatrix} 3 & -2 \\ -3 & 4 \end{bmatrix}$ [2] $\begin{bmatrix} 4 & 2 \\ 3 & 3 \end{bmatrix}$ [3] $6 \begin{bmatrix} 4 & 2 \\ 3 & 3 \end{bmatrix}$ [4] None of these
- Q.29** If A is a square matrix, then $A - A^T$ is
 [1] unit matrix [2] null matrix [3] A [4] a skew symmetric matrix
- Q.30** If $A = \begin{bmatrix} 1 & -2 & 3 \\ 4 & 0 & -1 \\ -3 & 1 & 5 \end{bmatrix}$, then $(\text{adj} A)_{23}$ is equal to
 [1] 13 [2] -13 [3] 5 [4] -5
- Q.31** Let A be a square matrix. Then which of the following is not a symmetric matrix
 [1] $A + A'$ [2] AA' [3] $A'A$ [4] $A - A'$
- Q.32** If A is a square matrix of order n, then the value of $|\text{adj} A|$ is
 [1] $|A|^{n-1}$ [2] $|A|^n$ [3] $|A|^{n+1}$ [4] $|A|^{n+2}$
- Q.33** If $A = \frac{1}{3} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ x & 2 & y \end{bmatrix}$ is orthogonal, then $x + y$ equals
 [1] 3 [2] 0 [3] -3 [4] 1

Q.34 The scalar matrix is

[1] $\begin{bmatrix} -1 & 3 \\ 2 & 4 \end{bmatrix}$

[2] $\begin{bmatrix} 0 & 3 \\ 2 & 0 \end{bmatrix}$

[3] $\begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix}$

[4] None of these

Q.35 In the following diagonal matrix is

[1] $\begin{bmatrix} 0 & 3 \\ 4 & 0 \end{bmatrix}$

[2] $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \end{bmatrix}$

[3] $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

[4] $\begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix}$

Q.36 The matrix $\begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}$ is a

[1] non-singular

[2] Idempotent

[3] Nilpotent

[4] Orthogonal

Q.37 Matrix $\begin{bmatrix} 1 & 1 & 3 \\ 5 & 2 & 6 \\ -2 & -1 & -3 \end{bmatrix}$ is -

[1] Nilpotent

[2] Idempotent

[3] Scalar matrix

[4] None of these

Q.38 If $A = \begin{bmatrix} ab & b^2 \\ -a^2 & -ab \end{bmatrix}$, then A is

[1] Involutory matrix

[2] Idempotent matrix

[3] Nilpotent matrix

[4] none of these

Q.39 For any square matrix $A = [a_{ij}]$, $a_{ij} = 0$, when $i \neq j$, then A is

[1] unit matrix

[2] scalar matrix

[3] diagonal matrix

[4] none of these

Q.40 the adjoint of symmetric matrix is

[1] symmetric matrix

[2] skew - symmetric matrix

[3] diagonal matrix

[4] none of these

Q.41 If $\begin{bmatrix} \alpha & \beta \\ \gamma & -\alpha \end{bmatrix}$ is square root of I_2 , then α, β and γ will satisfy the relation

[1] $1 + \alpha^2 + \beta \gamma = 0$

[2] $1 - \alpha^2 + \beta \gamma = 0$

[3] $1 + \alpha^2 - \beta \gamma = 0$

[4] $-1 + \alpha^2 + \beta \gamma = 0$

Q.42 If $A = \begin{bmatrix} a & b \\ b & a \end{bmatrix}$, then $|A + A^T|$ equals

[1] $4(a^2 - b^2)$

[2] $2(a^2 - b^2)$

[3] $a^2 - b^2$

[4] $4 ab$

Q.43 If $A = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix}$, then A^2 is equal to

[1] $\begin{bmatrix} \cos 2\alpha & \sin 2\alpha \\ -\sin 2\alpha & \cos 2\alpha \end{bmatrix}$

[2] $\begin{bmatrix} \cos 2\alpha & -\sin 2\alpha \\ \sin 2\alpha & \cos 2\alpha \end{bmatrix}$

[3] $\begin{bmatrix} \sin 2\alpha & \cos 2\alpha \\ -\cos 2\alpha & \sin 2\alpha \end{bmatrix}$

[4] $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

EXERCISE - 2

Q.1 If $\begin{bmatrix} x & y \\ u & v \end{bmatrix}$ is symmetric matrix, then

[1] $x + v = 0$

[2] $x - v = 0$

[3] $y + u = 0$

[4] $y - u = 0$

Q.2 If $E(\theta) = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$ then $E(\alpha) E(\beta)$ is equal to

[1] $E(0^\circ)$

[2] $E(\alpha\beta)$

[3] $E(\alpha + \beta)$

[4] $E(\alpha - \beta)$

Q.3 If $A = [a_{ij}]$ is scalar matrix of order $n \times n$ such that $a_{ij} = k$ for all i , then $|A|$ equals

[1] nk

[2] $n + k$

[3] n^k

[4] k^n

Q.4 If $M = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}$ and $M^2 - \lambda M - I_2 = 0$, then λ equals

[1] -2

[2] 2

[3] -4

[4] 4

Q.5 If A is orthogonal matrix, then $|A|$ equals

[1] 0

[2] 1

[3] -1

[4] ± 1

Q.6 If $A - A' = 0$, then A' is

[1] orthogonal matrix

[2] symmetrix matrix

[3] skew-symmetric matrix

[4] triangular matrix

Q.7 If A is any skew-symmetric matrix of odd orders, then $|A|$ equals

[1] -1

[2] 0

[3] 1

[4] none of these

Q.8 The adjoint of the matrix $\begin{bmatrix} 1 & -2 & 3 \\ 0 & 2 & -1 \\ -4 & 5 & 2 \end{bmatrix}$ is

[1] $\begin{bmatrix} 9 & 19 & -4 \\ 4 & 14 & 1 \\ 8 & 3 & 2 \end{bmatrix}$

[2] $\begin{bmatrix} 9 & 4 & 8 \\ 19 & 14 & 3 \\ -4 & 1 & 2 \end{bmatrix}$

[3] $\begin{bmatrix} 9 & -19 & -4 \\ -4 & 14 & -1 \\ 8 & -3 & 2 \end{bmatrix}$

[4] none of these

Q.9 If $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $B = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$ then $(A + B)^2$ equals-

[1] $A^2 + B^2$

[2] $A^2 + B^2 + 2AB$

[3] $A^2 + B^2 + AB - BA$

[4] none of these

Q.10 If $A = \begin{bmatrix} 3 & -4 \\ 1 & -1 \end{bmatrix}$ then for every positive integer n , A^n is equal to

[1] $\begin{bmatrix} 1+2n & 4n \\ n & 1+2n \end{bmatrix}$

[2] $\begin{bmatrix} 1+2n & -4n \\ n & 1-2n \end{bmatrix}$

[3] $\begin{bmatrix} 1-2n & 4n \\ n & 1+2n \end{bmatrix}$

[4] None of these

Q.11 If $A = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ and $(aI_2 + bA)^2 = A$, then

[1] $a = b = \sqrt{2}$

[2] $a = b = 1/\sqrt{2}$

[3] $a = b = \sqrt{3}$

[4] $a = b = 1/\sqrt{3}$

Q.12 If A be a matrix such that. inverse of 7A is the matrix $\begin{bmatrix} -1 & 2 \\ 4 & -7 \end{bmatrix}$, then A equals

[1] $\begin{bmatrix} 1 & 2 \\ 4 & 1 \end{bmatrix}$

[2] $\begin{bmatrix} 1 & 4/7 \\ 2/7 & 1/7 \end{bmatrix}$

[3] $\begin{bmatrix} 1 & 4 \\ 2 & 1 \end{bmatrix}$

[4] $\begin{bmatrix} 1 & 2/7 \\ 4/7 & 1/7 \end{bmatrix}$

Q.13 If $D = \text{diag} (d_1, d_2, \dots, d_n)$, then D^n equals

[1] D

[2] $\text{diag} (d_1^{n-1}, d_2^{n-1}, \dots, d_n^{n-1})$

[3] $\text{diag} (d_1^n, d_2^n, \dots, d_n^n)$

[4] None of these

Q.14 If $\begin{bmatrix} 1 & -\tan\theta \\ \tan\theta & 1 \end{bmatrix} \begin{bmatrix} 1 & \tan\theta \\ -\tan\theta & 1 \end{bmatrix}^{-1} = \begin{bmatrix} a & -b \\ b & a \end{bmatrix}$, then

[1] $a = 1, b = 1$

[2] $a = \cos 2\theta, b = \sin 2\theta$

[3] $a = \sin 2\theta, b = \cos 2\theta$

[4] None of these

Q.15 If $D = \text{diag} (d_1, d_2, d_3, \dots, d_n)$, where $d_i \neq 0$ for all $i = 1, 2, \dots, n$, then D^{-1} is equal to

[1] D

[2] I_n

[3] $\text{diag} (d_1^{-1}, d_2^{-1}, \dots, d_n^{-1})$

[4] None of these

Q.16 If A and B are square matrices such that $AB = B$ and $BA = A$, then $A^2 + B^2$ is equal to

[1] 2AB

[2] 2BA

[3] A + B

[4] AB

Q.17 If $A = \begin{bmatrix} 1 & -2 \\ 3 & 0 \end{bmatrix}$, $B = \begin{bmatrix} -1 & 4 \\ 2 & 3 \end{bmatrix}$ and $ABC = \begin{bmatrix} 4 & 8 \\ 3 & 7 \end{bmatrix}$, then C equals-

[1] $\frac{1}{66} \begin{bmatrix} 54 & 110 \\ 3 & 11 \end{bmatrix}$

[2] $\frac{1}{66} \begin{bmatrix} -54 & -110 \\ 3 & 11 \end{bmatrix}$

[3] $\frac{1}{66} \begin{bmatrix} -54 & 110 \\ 3 & -11 \end{bmatrix}$

[4] none of these

Q.18 If $A = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$, then

[1] $\text{adj } A = A$

[2] $\text{adj } A = A^{-1}$

[3] $A^{-1} = -A$

[4] none of these

EXERCISE - 3

- Q.1** If for suitable matrices A, B; $AB = A$ and $BA = B$; then A^2 equals- **[RPET 98]**
- [1] I [2] A [3] B [4] 0
- Q.2** If A and B are matrices of order 3×2 and C is of order 2×3 , then which of the following matrices is not defined- **[RPET 98]**
- [1] $A^T + B$ [2] $A^T + B^T$ [3] $A^T + C$ [4] $B + C^T$
- Q.3** If $A = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}$, then $A(\text{adj } A)$ equals- **[RPET 98]**
- [1] $\begin{bmatrix} 10 & 0 \\ 0 & 10 \end{bmatrix}$ [2] $\begin{bmatrix} 0 & 10 \\ 10 & 0 \end{bmatrix}$ [3] $\begin{bmatrix} 10 & 1 \\ 1 & 10 \end{bmatrix}$ [4] none of these
- Q.4** If $A = \begin{bmatrix} 2 & 3 \\ 1 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 3 & 2 \\ 2 & 3 & 4 \end{bmatrix}$, then AB equals- **[RPET 98]**
- [1] $\begin{bmatrix} 8 & 15 & 16 \\ 5 & 9 & 10 \end{bmatrix}$ [2] $\begin{bmatrix} 8 & 5 \\ 15 & 9 \\ 16 & 10 \end{bmatrix}$ [3] $\begin{bmatrix} 8 & 5 \\ 15 & 9 \end{bmatrix}$ [4] none of these
- Q.5** If for a square matrix $A = [a_{ij}]$, $a_{ij} = i^2 - j^2$ then A is- **[RPET 99]**
- [1] unit matrix [2] zero matrix [3] symmetric matrix [4] skew symmetric matrix
- Q.6** If $A = \begin{bmatrix} i & 0 \\ 0 & i \end{bmatrix}$, then A^{40} is equal to- **[RPET 99]**
- [1] $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ [2] $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ [3] $\begin{bmatrix} -1 & 1 \\ 0 & -1 \end{bmatrix}$ [4] $\begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$
- Q.7** If $A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 4 & 9 \\ 1 & 8 & 27 \end{bmatrix}$, then $|\text{adj } A|$ is equal to- **[RPET 99]**
- [1] 12 [2] 144 [3] 72 [4] none of these
- Q.8** If $A = \begin{bmatrix} -1 & 2 & 3 \\ -2 & 3 & -1 \\ 3 & 1 & 2 \end{bmatrix}$ and I is the unit matrix of order 3 then $A^2 + 9I$ is equal to- **[RPET 99]**
- [1] 2A [2] 4A [3] 6A [4] none of these
- Q.9** The minimum number of zero's in an upper triangular matrix of order $n \times n$ is **[RPET 99]**
- [1] $\frac{n(n-1)}{2}$ [2] $\frac{n(n+1)}{2}$ [3] $\frac{n(n-1)(n+1)}{2}$ [4] none of these

Q.10 If $A = \begin{bmatrix} 1 & \sin \theta & 1 \\ -\sin \theta & 1 & \sin \theta \\ 1 & -\sin \theta & 1 \end{bmatrix}$ where $0 \leq \theta \leq 2\pi$, then **[RPET 99]**

- [1] $\text{Det.}(A) = 0$ [2] $\text{Det.}(A) \in (0, \infty)$ [3] $\text{Det.}(A) \in [0, 4]$ [4] $\text{Det.}(A) \in [2, \infty)$

Q.11 For any matrix A , AA^T is- **[RPET 2000]**

- [1] unit matrix [2] symmetric matrix [3] skew symmetric matrix [4] diagonal matrix

Q.12 If matrix A is of order $p \times q$ and matrix B is of order $r \times s$ then $A - B$ will exist if **[RPET 2000]**

- [1] $p = q$ [2] $p = r, q = s$ [3] $p = q, r = s$ [4] $p = s, q = r$

Q.13 If $2A - \begin{bmatrix} 1 & 2 \\ 7 & 4 \end{bmatrix} = \begin{bmatrix} 3 & 0 \\ 0 & -2 \end{bmatrix}$, then A is equal to **[RPET 2000]**

- [1] $\begin{bmatrix} 2 & 2 \\ 7/2 & 1 \end{bmatrix}$ [2] $\begin{bmatrix} 4 & 4 \\ 7/2 & 1 \end{bmatrix}$ [3] $\begin{bmatrix} 3 & -1 \\ 7 & 2 \end{bmatrix}$ [4] None of these

Q.14 The inverse of the matrix $\begin{bmatrix} 4 & 7 \\ 1 & 2 \end{bmatrix}$ is- **[RPET 84, 96, 2001]**

- [1] $\begin{bmatrix} 2 & -7 \\ -1 & 4 \end{bmatrix}$ [2] $\begin{bmatrix} 2 & -1 \\ -7 & 4 \end{bmatrix}$ [3] $\begin{bmatrix} -2 & 7 \\ 1 & -4 \end{bmatrix}$ [4] none of these

Q.15 For any square matrix A , $A + A^T$ is- **[RPET 2001]**

- [1] unit matrix [2] symmetric matrix [3] skew symmetric matrix [4] zero matrix

Q.16 For two matrices A and B , if $AB = 0$, then **[RPET 2001]**

- [1] $A = 0$ and $B = 0$ [2] $A = 0$ or $B = 0$
[3] it is not necessary that $A = 0$ or $B = 0$ [4] all above are false

Q.17 If $A = \begin{bmatrix} 3 & -5 \\ -1 & 0 \end{bmatrix}$, then $\text{adj } A$ is equal to- **[RPET 2001]**

- [1] $\begin{bmatrix} 3 & 5 \\ 1 & 0 \end{bmatrix}$ [2] $\begin{bmatrix} 0 & -5 \\ -1 & 3 \end{bmatrix}$ [3] $\begin{bmatrix} 0 & 5 \\ 1 & 3 \end{bmatrix}$ [4] $\begin{bmatrix} 0 & -1 \\ -5 & 3 \end{bmatrix}$

Q.18 The inverse of $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ is **[RPET 2001]**

- [1] $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ [2] $\begin{bmatrix} 1 & -1 \\ 0 & 1 \end{bmatrix}$ [3] $\begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix}$ [4] $\begin{bmatrix} -1 & 1 \\ 0 & 1 \end{bmatrix}$

Q.19 If $A = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix}$, then A^2 equals- **[RPET 2001]**

- [1] $\begin{bmatrix} \cos 2\alpha & -\sin 2\alpha \\ \sin 2\alpha & \cos 2\alpha \end{bmatrix}$ [2] $\begin{bmatrix} \cos 2\alpha & \sin 2\alpha \\ -\sin 2\alpha & \cos 2\alpha \end{bmatrix}$ [3] $\begin{bmatrix} \sin 2\alpha & \cos 2\alpha \\ -\cos 2\alpha & \sin 2\alpha \end{bmatrix}$ [4] $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

Q.20 If $A = \begin{bmatrix} 1 & -2 & 1 \\ 2 & 1 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 \\ 3 & 2 \\ 1 & 1 \end{bmatrix}$, then $(AB)'$ equals **[RPET 2001]**

[1] $\begin{bmatrix} -3 & -2 \\ 10 & 7 \end{bmatrix}$

[2] $\begin{bmatrix} -3 & 10 \\ -2 & 7 \end{bmatrix}$

[3] $\begin{bmatrix} -3 & 10 \\ 7 & -2 \end{bmatrix}$

[4] none of these

Q.21 If A is a square matrix, then $A(\text{adj } A)$ equals- **[RPET 2002]**

[1] $|A|$

[2] I

[3] $|A| I$

[4] 0

Q.22 If $A = \begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix}$, then $A^2 - 5A$ equals- **[RPET 2002]**

[1] $7I_2$

[2] $-7I_2$

[3] $5I_2$

[4] $-5I_2$

Q.23 If $A = \begin{bmatrix} 4 & 2 \\ -1 & 1 \end{bmatrix}$ then $(A - 2I)(A - 3I)$ equals- **[RPET 2002]**

[1] 0

[2] A

[3] I

[4] $5I$

Q.24 If a $n \times n$ square matrix A is such that $|A| = D$ and $|\text{adj } A| = D'$, then **[RPET 2002]**

[1] $DD' = D^n$

[2] $DD' = D^{n-1}$

[3] $DD' = D^2$

[4] none of these

Q.25 If for a matrix A , $A^3 = I$, then A^{-1} is equal to- **[RPET 2002]**

[1] A

[2] A^2

[3] A^3

[4] none of these

Q.26 If $A = \begin{bmatrix} a & 0 \\ 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 5 & 1 \end{bmatrix}$, then value of a for which $A^2 = B$ is- **[IIT (Scr) 2002]**

[1] 1

[2] -1

[3] 4

[4] no real number

Q.27 If $A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 2 \\ 2 & 0 & 1 \end{bmatrix}$, then $|\text{adj } A|$ is equal to- **[RPET 2003]**

[1] 9

[2] 18

[3] 81

[4] 27

Q.28 For two suitable order matrices A, B ; correct statement is- **[RPET 2003]**

[1] $(AB)^{-1} = A^{-1}B^{-1}$

[2] $(AB)^{-1} = B^{-1}A^{-1}$

[3] $(AB)^{-1} = (BA)^{-1}$

[4] none of these

Q.29 If $A = \begin{bmatrix} a & b \\ b & a \end{bmatrix}$ and $A^2 = \begin{bmatrix} \alpha & \beta \\ \beta & \alpha \end{bmatrix}$ then **[AIEEE 2003]**

[1] $\alpha = a^2 + b^2, \beta = a^2 - b^2$

[2] $\alpha = a^2 + b^2, \beta = ab$

[3] $\alpha = a^2 + b^2, \beta = 2ab$

[4] $\alpha = 2ab, \beta = a^2 + b^2$

Q.30 If $A = \begin{bmatrix} \alpha & 2 \\ 2 & \alpha \end{bmatrix}$ and $|A^3| = 125$, then α is equal to- **[IIT (Scr) 2004]**

[1] 0

[2] ± 5

[3] ± 2

[4] ± 3

EXERCISE # 1

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
Ans.	4	1	3	2	1	3	3	2	1	1	4	4	1	4	3	2	2	2	1	1	4	3	4	4	1
Que.	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43							
Ans.	1	2	2	4	1	4	1	3	3	4	2	1	3	3	1	4	1	1							

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
Ans.	4	3	4	4	4	2	2	1	1	2	2	4	3	2	3	3	2	2

Que.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25
Ans.	2	1	1	1	4	1	2	4	1	3	2	2	1	1	2	3	3	2	2	2	3	2	1	1	2
Que.	26	27	28	29	30	31	32	33	34	35	36	37													
Ans.	4	3	2	3	4	2	4	3	1	1	4	1													